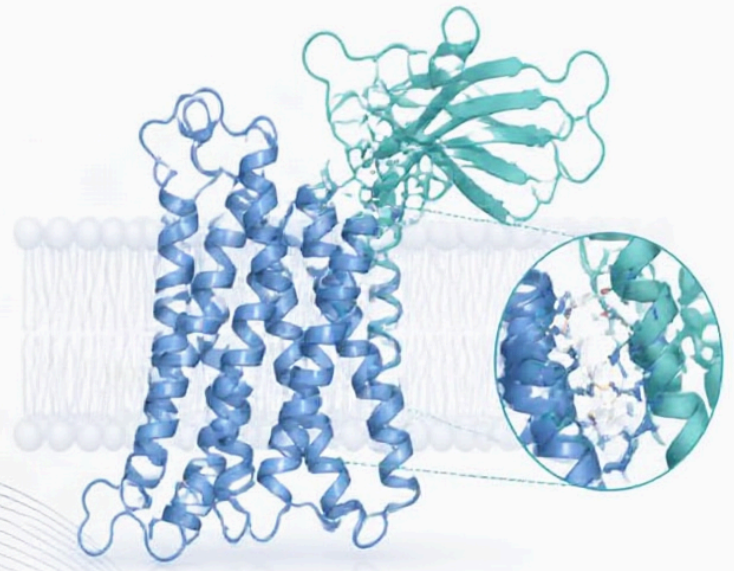


Structural Analysis of TMEM106B

Protein–Protein Interaction

An *in-silico* structural investigation using
PDB 9GI8 and PyMOL



Structural Analysis of TMEM106B

Protein–Protein Interaction

An in-silico structural investigation using PDB 9GI8 and PyMOL

Research Question

Does the experimentally determined structure of TMEM106B show a defined protein–protein interface between its two chains?

Why TMEM106B?

“TMEM106B (Transmembrane Protein 106B) was selected as a candidate protein for structural investigation within our broader anxiety-related bioinformatics project. Structural analysis was performed to determine whether an experimentally resolved TMEM106B structure contains a defined protein–protein interface that can be characterized computationally.”

Structure

PDB	9GI8
Protein	TMEM106B
Organism	Homo sapiens
Chains	A + B
Software	PyMOL
Analysis	Protein-protein interface

Workflow

PDB 9GI8 → Identify chains A & B	4 Å interface → Identify interface residues	3.5 Å close-contact analysis → Structural interpretation
-------------------------------------	--	---

Results

Metric	Chain A	Chain B
4 Å contact atoms	547	499
Residue-expanded interface	592	591
CA-defined interface residues	38	38
3.5 Å close-contact atoms	506	466
CA-defined close residues	29	26

Figure

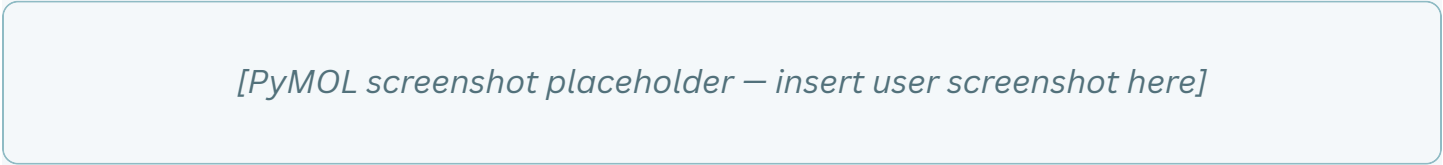
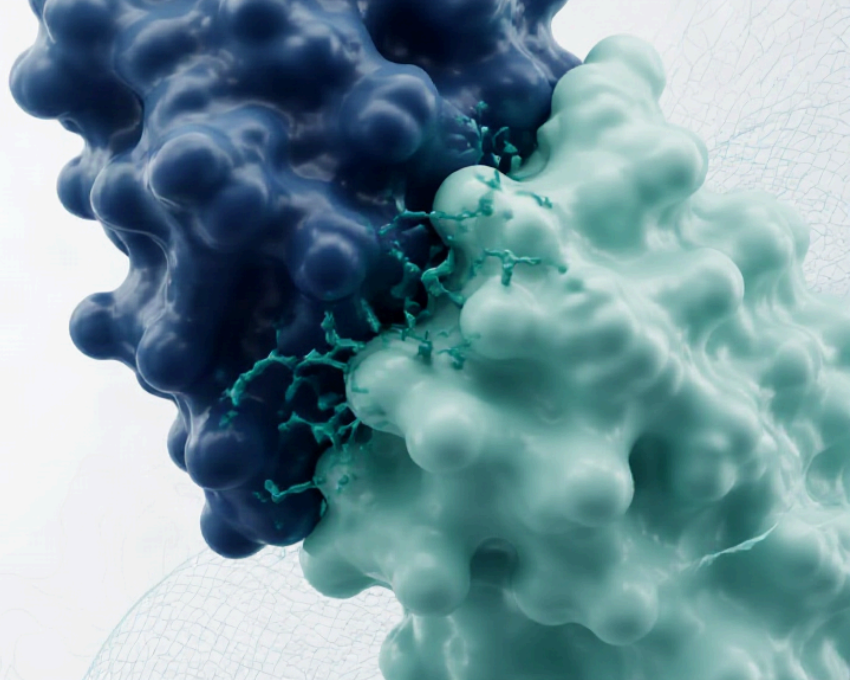


Figure 1. TMEM106B structure (PDB 9GI8) showing the analyzed chain A-chain B interface in PyMOL.

Interface Residues

Structural validation,
interpretation, and limitations



Interface Residues

Interface analysis identified residues approximately within the 54–92 region of the deposited chains. The interface was determined from three-dimensional proximity rather than residue-number matching.

Representative residues

THR54 • GLY55 • ARG56 • ASP57 • CYS61 • PRO62 • ARG69 • PRO71 • ARG72 • TYR84 • ASP86 • ARG88–92

Interface / Contact View

[PyMOL interface/contact screenshot placeholder – insert user screenshot here]

Interface/contact view from the user-provided PyMOL analysis.

Important Structural Validation

Same residue number ≠ same interaction

THR54(A) OG1 → THR54(B) OG1 = 40.440 Å

Identical residue numbering across chains does not imply interaction; 3D coordinates determine proximity.

What Did We Find?

The structural analysis of PDB 9GI8 demonstrates a substantial interface between the two TMEM106B chains. The presence of numerous inter-chain contacts within 4 Å, together with a smaller set of contacts within 3.5 Å, supports the presence of a defined structural association between chains A and B in the deposited model.

Relevance to Anxiety Project

This analysis represents the structural PPI component of the broader anxiety-related bioinformatics project. It provides structural information about TMEM106B and its chain-chain association. However, the structural result alone does not demonstrate that TMEM106B causes anxiety, contains an anxiety-causing mutation, or produces an anxiety phenotype. Such conclusions require additional genetic, transcriptomic, functional, and/or literature evidence.

Limitations

crowded distance visualization contained many simultaneous labels; exact shortest atom pair not reported as final quantitative result; short distance alone does not prove a specific H-bond/salt bridge; functional significance cannot be inferred from structural proximity alone.

Conclusion

PDB 9GI8 was successfully analyzed to characterize the structural interface between two TMEM106B chains. A 4 Å analysis identified 38 CA-defined interface residues on each chain, while the stricter 3.5 Å analysis identified 29 and 26 close-contact residues on chains A and B, respectively. These findings provide a reproducible structural PPI result that can be incorporated into the broader anxiety-related bioinformatics investigation. Further analyses are required to establish the functional and disease-related significance of TMEM106B.